

STONEWALL BANK BUOY 46050, USA

WMO: 996760

Lat: 44.677N

Lon: 124.515W

Elev: 0

StdP: 14.70

Time Zone: -8.00 (W08)

Period: 02-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	38.2	41.0	N/A	N/A	N/A	N/A	N/A	N/A	39.0	50.2	36.9	50.5	13.3	90	0.717

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.2	63.5	N/A	62.7	N/A	62.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	11.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
36.1	31.9	26.6	DB	34.0	65.5	3.9	2.5	31.2	67.3	28.9	68.8	26.7	70.2	23.8	72.1	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.6	49.0	48.0	47.8	49.0	52.7	56.1	57.7	57.6	57.3	55.1	51.7	48.8
	DBStd	4.78	3.19	3.24	2.53	2.26	2.54	2.21	2.87	2.33	2.97	2.77	2.92	4.01
	HDD50	327	54	68	74	42	5	0	0	0	0	2	16	66
	HDD65	4527	495	476	533	479	382	266	226	229	232	308	400	501
	CDD50	1278	25	12	7	13	88	184	239	236	219	159	66	30
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	13.9	15.8	14.8	15.3	14.3	12.8	13.1	12.7	10.3	11.3	13.1	15.7	17.5
(15) Precipitation	PrecAvg	80.3	13.3	9.1	8.4	6.3	4.0	2.7	0.6	0.8	1.9	5.9	12.2	13.6
	PrecMax	114.8	21.6	20.4	14.7	10.9	7.3	5.5	2.2	3.8	7.8	10.6	23.2	27.0
	PrecMin	61.2	5.4	2.8	1.7	2.7	0.3	0.5	0.1	0.0	0.1	0.9	2.9	5.8
	PrecStd	14.5	4.7	5.3	3.2	2.4	1.9	1.4	0.5	0.9	1.7	2.6	5.2	4.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.5	56.7	54.6	57.7	61.0	62.3	64.4	64.2	64.7	63.1	59.1	58.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	55.1	54.6	52.8	54.3	58.0	60.8	63.4	63.1	63.3	61.1	57.5	56.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	54.0	53.2	51.9	52.8	57.1	60.0	62.5	62.2	62.3	59.9	56.5	55.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	53.1	51.9	51.2	51.9	56.2	59.3	61.8	61.3	61.5	59.0	55.5	53.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	3.7	3.7	3.6	3.2	2.7	2.8	3.2	3.6	3.7	3.5	3.9	4.2
		MCDBR	3.8	4.1	4.1	4.0	3.6	3.4	3.6	4.0	4.2	3.8	4.1	5.0
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.327	0.328	0.341	0.378	0.383	0.362	0.350	0.354	0.350	0.341	0.331	0.328
	taud		2.524	2.543	2.525	2.379	2.402	2.484	2.540	2.537	2.540	2.556	2.537	2.498
	Ebn at Noon		252	274	285	280	281	287	290	285	278	266	248	238
	Edh at Noon		22	25	29	36	36	34	31	30	28	24	21	20
(44) All-Sky Solar Radiation	RadAvg		423	691	1036	1479	1806	1942	2004	1686	1372	853	486	352
	RadStd		59	70	110	102	136	121	148	152	103	83	60	47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (23 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+31

Nomenclature: See separate page